

Assignment and Rename Operations in Relational Algebra

The Assignment Operation

Definition

The assignment operation ($\leftarrow$) allows assigning the result of a relational-algebra expression to a temporary relation variable. It is analogous to assignment in programming languages and helps break down complex queries.

Example: Find courses offered in both Fall 2017 and Spring 2018 semesters.

```
courses_fall_2017  $\leftarrow$   $\pi_{\text{course\_id}}(\sigma_{\text{semester}=\text{"Fall"} \wedge \text{year}=2017}(\text{section}))$   
courses_spring_2018  $\leftarrow$   $\pi_{\text{course\_id}}(\sigma_{\text{semester}=\text{"Spring"} \wedge \text{year}=2018}(\text{section}))$   
result = courses_fall_2017  $\cap$  courses_spring_2018
```

Explanation:

- The first two lines assign projected course IDs for Fall 2017 and Spring 2018 to temporary variables.
- The last line computes the intersection of these two sets.

The Rename Operation

Definition

The rename operation, denoted by the Greek letter ρ , assigns a new name to a relation or its attributes, enabling clearer queries and handling of multiple occurrences of the same relation.

Basic form:

$$\rho_x(E)$$

returns the result of expression E with the relation named x .

Extended form with attribute renaming: If E has attributes $A_1, A_2, \dots, A_n$, then

$$\rho_{x(B_1, B_2, \dots, B_n)}(E)$$

returns the result of E as relation x but renames attributes to $B_1, B_2, \dots, B_n$.

Use case: Rename is especially useful when the same relation appears multiple times in a query or when attributes need clearer names.

Example: Finding Instructors Earning More Than Instructor 12121

Suppose we want to find the *ID* and *name* of instructors earning more than the instructor with ID 12121.

$$\begin{aligned} \text{inst1} &\leftarrow \rho_{I_1}(\text{instructor}) \\ \text{inst2} &\leftarrow \rho_{I_2}(\text{instructor}) \\ \text{result} &= \pi_{\text{ID}, \text{name}}(\sigma_{I_1.\text{salary} > I_2.\text{salary} \wedge I_2.\text{ID} = 12121}(\text{inst1} \times \text{inst2})) \end{aligned}$$

Explanation:

- Rename the `instructor` relation twice to `inst1` and `inst2` for self-join.
- Perform Cartesian product `inst1` $\times$ `inst2`.
- Select tuples where `inst1.salary` is greater than `inst2.salary` and `inst2.ID` is 12121.
- Project only the `ID` and `name` attributes from `inst1`.